Eshara – Database Schema (ERD)

Logical Design (Main Tables)

Users & Authentication

AspNetUsers

Id	uniqueidentifier	PK
FullName	nvarchar(256)	
XP	int	
StreakDays	int	
LastActivityDate	datetime2	
CreatedAt	datetime2	
UserName	nvarchar(256)	
NormalizedUserName	nvarchar(256)	
Email	nvarchar(256)	
NormalizedEmail	nvarchar(256)	
EmailConfirmed	bit	
PasswordHash	nvarchar(max)	
SecurityStamp	nvarchar(max)	
ConcurrencyStamp	nvarchar(max)	
PhoneNumber	nvarchar(max)	
PhoneNumberConfirmed	bit	
TwoFactorEnabled	bit	
LockoutEnd	datetimeoffset	
LockoutEnabled	bit	
AccessFailedCount	int	

Many-to-Many (AspNetUsers ↔ AspNetRoles)

AspNetRoles

Id	uniqueidentifier	PK
Name	nvarchar(256)	
NormalizedName	nvarchar(256)	
ConcurrencyStamp	nvarchar(max)	

Logical view note

A user can hold many roles and a role can be assigned to many users. Physically this is resolved by the AspNetUserRoles junction table (see the Physical Design page).

Dictionary

ContentCategories

Id	uniqueidentifier	PK
Name	nvarchar(256)	
Description	nvarchar(max)	
CreatedAt	datetime2	

DictionaryItems

Id	uniqueidentifier	PK
Title	nvarchar(200)	
Description	nvarchar(2000)	
VideoStorageKey	nvarchar(500)	
CategoryId	uniqueidentifier	FK
IsPublished	bit	
CreatedAt	datetime2	
UpdatedAt	datetime2	

User Progress

UserDictionaryProgresses

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
DictionaryItemId	uniqueidentifier	FK
LearnedAt	datetime2	

Translation

TranslationHistories

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
Type	int	
InputText	nvarchar(4000)	
OutputText	nvarchar(4000)	
IsSuccessful	bit	
CreatedAt	datetime2	

Gamification

XpTransactions

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
SourceType	int	
TranslationHistoryId	uniqueidentifier	FK
UserDictionaryProgressId	uniqueidentifier	FK
Amount	int	
CreatedAt	datetime2	

Legend

- PK (Primary Key)
- FK (Foreign Key)
- Relationship (1 – Many)
- Many-to-Many (logical)

1 = one side of a relationship * = many side 0..1 = optional one

- Users & Authentication
- Dictionary
- Translation
- User Progress
- Gamification

Eshara – Database Schema (ERD)

Physical Design (Full Implementation)

Users & Authentication

AspNetUsers

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FullName	nvarchar(256)	
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LastActivityDate	datetime2	
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UserName	nvarchar(256)	
NormalizedUserName	nvarchar(256)	
Email	nvarchar(256)	
NormalizedEmail	nvarchar(256)	
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PasswordHash	nvarchar(max)	
SecurityStamp	nvarchar(max)	
ConcurrencyStamp	nvarchar(max)	
PhoneNumber	nvarchar(max)	
PhoneNumberConfirmed	bit	
TwoFactorEnabled	bit	
LockoutEnd	datetimeoffset	
LockoutEnabled	bit	
AccessFailedCount	int	

AspNetUserRoles

UserId	uniqueidentifier	PK, FK
RoleId	uniqueidentifier	PK, FK

AspNetRoles

Id	uniqueidentifier	PK
Name	nvarchar(256)	
NormalizedName	nvarchar(256)	
ConcurrencyStamp	nvarchar(max)	

AspNetUserClaims

Id	int	PK
UserId	uniqueidentifier	FK
ClaimType	nvarchar(max)	
ClaimValue	nvarchar(max)	

AspNetRoleClaims

Id	int	PK
RoleId	uniqueidentifier	FK
ClaimType	nvarchar(max)	
ClaimValue	nvarchar(max)	

AspNetUserLogins

LoginProvider	nvarchar(256)	PK
ProviderKey	nvarchar(256)	PK
ProviderDisplayName	nvarchar(max)	
UserId	uniqueidentifier	FK

AspNetUserTokens

UserId	uniqueidentifier	PK, FK
LoginProvider	nvarchar(256)	PK
Name	nvarchar(256)	PK
Value	nvarchar(max)	

RefreshTokens

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
TokenHash	nvarchar(max)	
ExpiresAt	datetime2	
CreatedAt	datetime2	
RevokedAt	datetime2	
ReplacedByTokenId	uniqueidentifier	FK

Dictionary

ContentCategories

Id	uniqueidentifier	PK
Name	nvarchar(256)	
Description	nvarchar(max)	
CreatedAt	datetime2	

DictionaryItems

Id	uniqueidentifier	PK
Title	nvarchar(200)	
Description	nvarchar(2000)	
VideoStorageKey	nvarchar(500)	
CategoryId	uniqueidentifier	FK
IsPublished	bit	
CreatedAt	datetime2	
UpdatedAt	datetime2	

User Progress

UserDictionaryProgresses

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
DictionaryItemId	uniqueidentifier	FK
LearnedAt	datetime2	

Translation

TranslationHistories

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
Type	int	
InputText	nvarchar(4000)	
OutputText	nvarchar(4000)	
IsSuccessful	bit	
CreatedAt	datetime2	

Gamification

XpTransactions

Id	uniqueidentifier	PK
UserId	uniqueidentifier	FK
SourceType	int	
TranslationHistoryId	uniqueidentifier	FK
UserDictionaryProgressId	uniqueidentifier	FK
Amount	int	
CreatedAt	datetime2	

Legend

- PK (Primary Key)
- FK (Foreign Key)
- Relationship (1 – Many)

Users & Authentication · Dictionary · Translation · User Progress · Gamification

1 = one side of a relationship * = many side 0..1 = optional one